
Experiment Title

Implementation of Boolean Expression $F = P \oplus Q \oplus R$ using Raspberry Pi Pico 2W

Components Required

- Raspberry Pi Pico 2W
- Breadboard
- LED
- Jumper Wires
- USB Cable
- Arduino IDE

Connections

- LED Long Leg (+) $\rightarrow$ GP1
- LED Short Leg (-) $\rightarrow$ GND

Procedure

1. Place Raspberry Pi Pico 2W on the breadboard.
2. Connect the long leg of the LED to GP1.
3. Connect the short leg of the LED to GND.
4. Open Arduino IDE.
5. Select Raspberry Pi Pico 2W board.
6. Create a new sketch and upload the program.
7. Observe the LED output after uploading.

Observation

For the Boolean expression $F = P \oplus Q \oplus R$, the output becomes HIGH for the selected input combination, and the LED turns ON.

Result

The Boolean expression $F = P \oplus Q \oplus R$ was successfully implemented using Raspberry Pi Pico 2W, and the output was verified using an LED. The LED turned ON when the output was HIGH.